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System and method for biological methane gas generation and removal of carbon dioxide therefrom

Abstract

A system for biological methane production and removing carbon dioxide from the methane comprises (a) a primary anaerobic digester adapted and arranged to generate a gaseous mixture comprising methane and carbon dioxide from organic materials; (b) an electrochemical reactor comprising at least one reactor cell including an anode spaced from a cathode by a porous ion exchange resin wafer with a cation exchange membrane between the anode and the resin wafer and a bipolar ion exchange membrane between the cathode and the resin wafer; the electrochemical reactor being adapted and arranged to capture gaseous carbon dioxide within the resin wafer as aqueous bicarbonate, and to electrochemically generate hydrogen gas at the cathode; and (c) a hydrogenotrophic methanogenesis bioreactor adapted and arranged to convert the bicarbonate and hydrogen from the electrochemical reactor to methane. An electrochemical reactor and a method for producing methane with reduced carbon dioxide content also are described.

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Renewable Biological Systems For Alternative Sustainable Energy Production (FAO Agricultural

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Background/Summary

FIELD OF THE INVENTION

(1) This invention relates to a system and method for generating methane from biological waste materials and removing carbon dioxide from the methane using an electrochemical apparatus.

BACKGROUND OF THE INVENTION

(2) Efficient capture and removal of carbon dioxide from biologically generated methane gas is an essential technology for the development of more environmentally acceptable waste-based energy production systems. Converting carbon dioxide produced in wastewater treatment to energy is a strategy for ensuring water and energy resilience. Using renewable resources such as solar and wind for carbon dioxide utilization is also an important strategy to reduce the impact of greenhouse gasses on climate change. Biological conversion of organic materials such as food waste, wastewater and waste treatment sludge can use used on a large scale. For example, carbon dioxide produced in anaerobic digesters and flue gasses can be converted to methane in hydrogenotrophic methanogenesis bioreactors.

(3) Although microbial populations in bioreactors are robust with respect to toxic compounds in the feed streams there are still challenges in anaerobic gas fermentation to produce methane, including (1) the low aqueous solubility of carbon dioxide and (2) obtaining a suitable, convenient and cost effective source of hydrogen. The low solubility of carbon dioxide leads to low carbon dioxide utilization in bioreactors operated in open systems as atmospheric pressure. The source, production cost and storage of hydrogen frequently makes the economics and investment potential of biological methane production unfavorable.

(4) Because of the aforementioned challenges there is an ongoing need for alternative processes for biological production of methane from organic/biological waste streams.

SUMMARY OF THE INVENTION

(5) A system and method for biological methane gas production and removing carbon dioxide from the methane is described herein. The system comprises (a) a primary anaerobic digester adapted and arranged to generate a gaseous mixture comprising methane and carbon dioxide from organic materials; (b) an electrochemical reactor comprising at least one reactor cell including an anode spaced from a cathode by a porous ion exchange resin wafer with a cation exchange membrane between the anode and the resin wafer and a bipolar ion exchange membrane between the cathode and the resin wafer; the electrochemical reactor being adapted and arranged to capture gaseous carbon dioxide within the resin wafer as aqueous bicarbonate, and to electrochemically generate hydrogen gas at the cathode; and (c) a hydrogenotrophic methanogenesis bioreactor adapted and arranged to convert the bicarbonate and hydrogen from the electrochemical reactor to methane. The primary anaerobic digester includes a gas transfer line in fluid communication with an inlet end of the electrochemical reactor adapted and arranged to receive gas and liquid into pores of the resin wafer. The electrochemical reactor further includes a first gas vent line; an outlet operably connected to a liquid transfer line in fluid communication with the hydrogenotrophic

methanogenesis bioreactor; and a hydrogen transfer line providing fluid communication between the hydrogenotrophic methanogenesis bioreactor and a region adjacent to the cathode where hydrogen is produced during use. The hydrogenotrophic methanogenesis bioreactor comprises: a liquid recycle line in fluid communication with the inlet end of the electrochemical reactor adapted and arranged to direct fluid flow through the electrochemical reactor; and a second gas vent line.

(6) In use, a gaseous mixture containing methane and carbon dioxide is formed from the organic materials in the primary anaerobic digester. The electrochemical reactor is filled with an aqueous fluid, and the gaseous mixture from the primary anaerobic digester is transferred into the inlet end of the electrochemical reactor via the gas transfer line to flow through the resin wafer while a voltage is applied across the cathode and the anode of the electrochemical reactor. The carbon dioxide from the gaseous mixture is electrochemically converted to bicarbonate in the aqueous fluid within electrochemical reactor; while hydrogen simultaneously is electrochemically generated at the cathode. The methane from the gaseous mixture and the aqueous fluid both flow through the resin wafer, and carbon dioxide-depleted methane is vented through the first gas vent line and stored for later use. The aqueous fluid containing the bicarbonate flows through the liquid transfer line into the hydrogenotrophic methanogenesis bioreactor, and the hydrogen generated at the cathode flows through the hydrogen transfer line into the hydrogenotrophic methanogenesis bioreactor. The bicarbonate and hydrogen are biologically converted to methane in the hydrogenotrophic methanogenesis bioreactor, and the methane is vented through the second gas vent line and collected for later use. Aqueous fluid is recycled back into the inlet end of the electrochemical reactor from the hydrogenotrophic methanogenesis bioreactor through the liquid recycle line to maintain a flow of liquid within the electrochemical reactor when in use.

(7) The tandem electrochemical and gas-phase bioreactor system and method described herein provide a number of advantages for biological methane production. For example, the system generates high-purity renewable methane rather than just biogas (a mixture of methane and carbon dioxide). In addition, utilizing electrochemically produced hydrogen and carbon dioxide captured as aqueous bicarbonate in a hydrogenotrophic methanogenesis bioreactor boosts the overall methane production by at least about 40% and reduces capital and production costs by at least about 20% compared to traditional anaerobic digestion processes.

(8) The primary anaerobic digester of system can comprise a dual reactor that includes (i) an acidogenesis reactor adapted and arranged to biologically convert organic waste and wastewater to soluble volatile fatty acids (VFA), methane, carbon dioxide, and hydrogen; and (ii) an acetoclastic methanogenesis reactor adapted and arranged to biologically convert the VFA to carbon dioxide and methane. The gas transfer line of the dual reactor is in fluid communication with both the acidogenesis reactor and the acetoclastic methanogenesis reactor; and the acidogenesis reactor is in fluid communication with the acetoclastic methanogenesis reactor via a liquid feed line.

(9) In use, VFA, carbon dioxide, hydrogen and methane are generated in the acidogenesis reactor, and VFA-containing liquid from the acidogenesis reactor is fed into the acetoclastic methanogenesis reactor via the liquid feed line. Methane and carbon dioxide are generated in the acetoclastic methanogenesis reactor from the VFA, and carbon dioxide, methane, and hydrogen are transferred from the acidogenesis reactor and the acetoclastic methanogenesis reactor into the inlet end of electrochemical reactor through the gas transfer line.

(10) A preferred electrochemical reactor, also referred to as “electrochemical reactor for carbon and hydrogen delivery” (ERCHD), comprises a plurality (e.g., 2 to about 100) of the electrochemical reactor cells in a stack with the cathode of each adjacent cell separated from the anode of the nearest adjacent cell by an electrical insulator; and the reactor cells are in fluid communication with each other in series so that liquid and gas flow through each resin wafer sequentially. Each wafer can have a thickness in the range of about 1 to about 20 mm.

(11) Optionally, the electrochemical reactor includes at least one a porous gas and liquid flow distributor in contact with the resin wafer(s). The flow distributor is adapted and arranged to

laterally distribute gas and liquid within pores thereof prior to entering the resin wafer of the electrochemical reactor. Preferably, the flow distributor includes interconnected pores having an average pore size in the range of about 100 to about 600 micrometers, and is adapted and arranged to laterally distribute the liquid and gas bubbles having an average diameter in the range of about 100 to about 600 micrometers throughout the flow distributor and into the resin wafer of the electrochemical reactor. Optionally, flue gas can be introduced into the electrochemical reactor along with the gases from the anaerobic digester to supplement the carbon dioxide from the digester.

(12) A method for biologically generating carbon dioxide-depleted biogas comprises the steps of: (a) generating a biogas comprising methane and carbon dioxide by anaerobic degradation of biological material in an anaerobic digester; (b) passing the biogas generated in step (a) through an electrochemical reactor comprising an anode spaced from a cathode by a porous ion exchange resin wafer with a cation exchange membrane between the anode and the resin wafer and a bipolar ion exchange membrane between the cathode and the resin wafer, while applying a voltage across the anode and the cathode to convert carbon dioxide in the biogas to bicarbonate and to generate hydrogen gas by electrochemical hydrogen evolution at the cathode from protons produced by an electrochemical water splitting reaction occurring on the bipolar membrane surface; (c) venting and collecting the methane from the biogas that passes through the electrochemical reactor; (d) passing the hydrogen and bicarbonate formed in step (b) into a hydrogenotrophic methanogenesis bioreactor; (e) generating methane from the hydrogen and bicarbonate in the hydrogenotrophic methanogenesis bioreactor; and (f) venting and collecting the methane that forms in the hydrogenotrophic methanogenesis bioreactor.

(13) In some embodiments of the method, the anaerobic digester comprises: (i) an acidogenesis reactor adapted and arranged to biologically convert organic waste and wastewater to soluble volatile fatty acids (VFA), methane, carbon dioxide, and hydrogen; and (ii) an acetoclastic methanogenesis reactor adapted and arranged to biologically convert the VFA to carbon dioxide and methane. In use, VFA, carbon dioxide, hydrogen and methane are generated in the acidogenesis reactor; VFA-containing liquid from the acidogenesis reactor is fed into the acetoclastic methanogenesis reactor; methane and carbon dioxide are generated in the acetoclastic methanogenesis reactor, and carbon dioxide, methane, and hydrogen are transferred from the acidogenesis reactor and the acetoclastic methanogenesis reactor into electrochemical reactor.

(14) The systems, electrochemical reactors, and methods described herein comprise certain novel features and a combination of parts hereinafter fully described, which are illustrated in the accompanying drawings, and particularly pointed out in the appended claims. It is to be understood that various changes in the details may be made without departing from the spirit, or sacrificing any of the advantages of the systems, electrochemical reactors, and methods described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A provides a schematic illustration of a system for biologically generating a methane-containing biogas and removing carbon dioxide from the biogas as bicarbonate, comprising a primary anaerobic digester connected to an electrochemical reactor for removing carbon dioxide from the biogas and electrochemically generating hydrogen, and a hydrogenotrophic methanogenesis bioreactor for converting the hydrogen and bicarbonate into additional methane.

(2) FIG. 1B provides a more detailed illustration of an embodiment of the electrochemical reactor of the system in FIG. 1A.

(3) FIG. 2 is a schematic illustration of an alternative embodiment of an electrochemical reactor.

(4) FIG. 3 provides a flow diagram of a method for generating biological methane with reduced

carbon dioxide content

(5) FIG. 4 schematically illustrates an apparatus for evaluating carbon dioxide removal from a gas stream using an electrochemical reactor.

(6) FIG. 5 provides plots of percentage carbon dioxide captured and concentration of hydrogen produced versus operation time, for the apparatus of FIG. 4.

(7) FIG. 6 provides plots of carbon dioxide captured in milliequivalents (meq) and hydrogen produced (meq) versus operation time, for the apparatus of FIG. 4.

DETAILED DESCRIPTION OF THE PRESENT INVENTION

(8) The organic fraction of municipal solid waste comprises more than 50% of the almost 300 million tons of municipal solid waste generated in the U.S. per year. This fraction is difficult to recycle, and the majority of this waste is landfilled. An additional 55 million dry tons of water resource recovery facility residuals and manure slurries are produced per year. These potential energy sources are largely untapped as waste-to-energy technologies are currently expensive to build and demanding to operate and maintain. The increase in efficiency and reduction in cost resulting from the successful development of the systems and methods described herein can substantially reduce the gasoline gallon equivalent of renewable methane and propel waste-produced renewable energy to become financially favorable.

(9) The systems and methods described herein couple biological methanogenesis and electrochemical conversion of carbon dioxide and hydrogen generation to efficiently form high purity renewable methane at a significantly methane higher production rate than conventional, state of the art biological methanation processes. The systems and methods utilize three processing stages including anaerobic digestion of organic materials (e.g., wastewater, sludge, food waste, etc.) to produce a biogas, an electrochemical reactor to capture and remove carbon dioxide from the biogas for form aqueous bicarbonate, while simultaneously electrochemically generating hydrogen, and finally hydrogenotrophic methanogenesis to convert the bicarbonate and hydrogen into more methane. Overall, by combining carbon dioxide capture with hydrogen production, the systems and methods described herein is estimated to boost the overall methane production rate from anaerobic digestion of organic matter by up to 40% while reducing production costs by at least about 20%.

(10) Anaerobic digestion involves hydrolysis, acidogenesis, acetogenesis, and methanogenesis of organic materials (e.g., food waste, wastewater, waste-treatment sludge, and the like). Hydrolysis involves solubilization of particulates and depolymerization of complex materials such as proteins and carbohydrates. Acidogenesis is a process in which the hydrolyzed materials are converted to volatile fatty acids (low molecular weight organic acids). Acetogenesis is a process in which the volatile fatty acids are converted to acetic acid, carbon dioxide and hydrogen. Methanogenesis is a process the utilizes the acetic acid and hydrogen from the acetogenesis process to produce methane and carbon dioxide. These four stages typically occur together in an anaerobic digester.

(11) Anaerobic digestion systems can be separated into an acetogenesis stage and a methanogenesis stage. In acetogenesis, acidogenic microorganisms (acidogens) and hydrogen-producing microorganisms generate hydrogen and volatile fatty acids (VFAs). In methanogenesis, acetogenic microorganisms (acetogens) and methane producing microorganisms (methanogens) convert the VFAs into methane (see, e.g., S. Li, X. Yang, in Handbook of Biofuels Production (Second Edition), 2016).

(12) Hydrolysis involves bacteria such as bacterial form the genera *Bacillus*, *Cellulomonas* and *Eubacterium*. Acidogenesis involves bacterial from genera such as *Propionibacterium*, *Butyrivibrio* and *Acetivibrio*. Acetogenesis involves bacteria from genera such as *Clostridia* and *Acetivibrio*, in particular *Clostridium aceticum*, *Acetobacter woodii* and *Clostridium termotrophicum*.

Methanogenesis involves microorganisms of the order Archaea, which are phylogenetically distinct from both eukaryotes and bacteria, but are often found in association with anaerobic bacteria.

Methanogens belong to the phylum Euryarchaeota in five orders that include Methanobacteriales, Methanococcales, Methanomicrobiales, Methanopyrales, and Methanosarcinales. Methanogens do

not utilize oxygen for respiration. Instead, methanogens use carbon as the terminal electron acceptor in methanogenesis. The carbon generally comes from low molecular weight organic compounds. The two well described pathways for methanogenesis involve the use of acetic acid or inorganic carbon dioxide as terminal electron acceptors (see Equations (Eq.) 1 and 2):

$\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$ Eq. 1:

$\text{CH}_3\text{COOH} \rightarrow \text{CH}_4 + \text{CO}_2$ Eq. 2:

(13) Acetogenesis can be illustrated by Equation 3:

$2\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_3\text{COOH} + 2\text{H}_2\text{O}$ Eq. 3.

(14) There are three pathways for methanogenesis. Acetoclastic methanogenesis converts acetate to methane and carbon dioxide by Equation 2 described above.

(15) Acetoclastic methanogens include the Methanosarcinales, such as *Methanosarcina* and *Methanosaeta*. Methylophilic methanogenesis utilizes methanol and methylamine as the source of carbon for methane production. Hydrogenotrophic methanogenesis, which occurs in all five of the Euryarchaeota methanogenic orders described above, use hydrogen for the reduction of carbon dioxide according to Equation 1 described above, or reduction of CO or formate by similar processes.

(16) Anaerobic digesters such as acetogenic reactors and methanogenic reactors are described, e.g., in United Nations Food and Agriculture Organization (FAO) Agricultural Services Bulletin-128, "Renewable biological systems for alternative sustainable energy production", Chapter 4, 1997, available online at the fao.org website; which is incorporated herein by reference in its entirety.

(17) In the electrochemical reactors described herein, carbon dioxide is captured as bicarbonate by reaction with electrochemically produced hydroxide in the resin wafers. Preferably, in use, the pH of fluid within the resin wafer of the electrochemical reactor is maintained in the range of about 7.5 to about 9.5 by the electric potential applied across the anode and the cathode. Typically, the electric potential applied across the anode and cathode is at least above 1.23 Volt (e.g., about 1.23 to about 5 Volts) per reactor cell for a desired current. The actual voltage per reactor cell pair will vary with the thickness of the resin wafer, as will be evident to those of ordinary skill in the art. Thicker wafers will require higher voltages. Preferably, the voltage per cell pair will be greater than 1.23 V if the wafer thickness is greater than 1 mm. Typically, the wafers will have a thickness in the range of about 1 mm to about 20 mm. However, greater wafer thickness may still work. The determination of an appropriate working voltage is within the level of ordinary skill in the art.

(18) In some preferred embodiments, a porous gas distributor is positioned across the inlet end of each wafer. The distributor comprises a porous polymeric foam or sintered material with interconnected pores positioned in close contact with or integral with the inlet end of each wafer, such that liquid and gas entering the distributor are dispersed laterally through the porous material of the distributor before entering the resin wafer. The lateral dispersion of the gas in the porous distributor provides an even plug-flow distribution of the gas and fluid, which reduces gas channeling in the wafers during operation of the electrochemical reactor.

(19) The porous gas and liquid distributors ("porous distributors") can be included as separate pieces that are arranged in close contact across the inlet end of the wafer. Alternatively, the porous distributors can be integral with (e.g., fused to) the wafers. In preferred embodiments, the porous distributors are formed from a porous polymeric foam (e.g., polyurethane or polyethylene foam). Each porous distributor is adapted for introducing CO₂-containing gas bubbles into an aqueous fluid within the resin wafer to direct an even distribution of micro-sized gas bubbles into the resin wafer to facilitate conversion of CO₂ from the biogas into bicarbonate ion. The aqueous fluid and gas are laterally dispersed and mixed together as they pass through the interconnected pores of the distributor and into the wafer. The porous distributor disperses the gas into micro-sized bubbles, and creates an even "plug-flow" distribution in the aqueous fluid carrier entering the wafer.

(20) As used herein, the terms "micro-sized gas bubbles", "microbubbles", and grammatical

variations thereof, refer to gas bubbles having a diameter of less than about 1000 micrometers. The size (i.e., volume) of the pores of the gas distributor influences the size of gas bubbles produced therein. Preferably, the porous distributor has an average pore diameter of about 100 to about 600 micrometers. While it is understood that the pores of the gas distributor may not be spherical in shape, for convenience, the phrase “average pore size of about 100 micrometers” refers to pores having a volume equivalent to the volume of a sphere having a diameter of 100 micrometers. Such gas and liquid flow distributors are described, e.g., in co-owned U.S. Pat. No. 9,339,764 to YuPo J. Lin et al., which is incorporated herein by reference in its entirety.

(21) The electrochemical reactor also is adapted to vent CO₂-depleted methane-containing gas from the reactor. Gas can be introduced into or vented from the wafers or portions thereof via gas inlet and/or outlet tubes, via microporous membranes or channels in contact with or connected to the wafers or portions, or by any other suitable structure. The cathodes, resin wafers with their accompanying gas distributors, and anodes are interleaved with alternating cation exchange and bipolar membranes and collectively direct a net flow of protons through the wafers toward the direction of the cathode and a net flow of hydroxyl ions through the wafers toward the direction of the anode when an electric potential is applied to across cathode and anode. The flow of protons and hydroxyl ions, combined with the capture of CO₂, maintains a basic pH in each basic wafer or basic portion and an acidic pH in each acidic wafer or acidic portion. Hydrogen gas is electrochemically generated at the cathode aided by water splitting facilitated by the bipolar membrane. Anodes and cathodes of adjacent reactor cells are separated by an electrical insulator such as a non-conducting polymer.

(22) Porous solid resin wafer ion exchange materials suitable for use in the present invention can be prepared in any suitable manner known in the art, such as for example the materials and methods described in U.S. Pat. No. 6,495,014 (Datta et al., incorporated herein by reference in its entirety), U.S. Pat. No. 7,452,920 (Lin et al., incorporated herein by reference in its entirety), and U.S. Pat. No. 7,306,934 (Arora et al.). Commercial resin wafers are also available than can be used or adapted for use in the electrochemical reactors described herein.

(23) FIG. 1A provides a schematic illustration of a system for biologically generating a methane-containing biogas and removing carbon dioxide from the biogas as bicarbonate, generating hydrogen, and converting the bicarbonate and hydrogen into methane. System **100** comprises a two-stage primary anaerobic digester system **102** including acidogenesis reactor **104** and acetoclastic methanogenesis reactor **106**. A liquid feed line **108** provides for transfer of VFA-containing liquid from reactor **104** to reactor **106**. Acidogenesis reactor **104** includes a gas transfer line **110** for transferring gases generated in reactor **104** into ERCHD electrochemical reactor **112**. Acetoclastic methanogenesis reactor **106** also includes a gas transfer line **124** for transferring gases generated in reactor **106** into ERCHD electrochemical reactor **112**. Reactor **112** includes a gas vent **114** to vent methane for storage and later use, a liquid transfer line **116** and hydrogen transfer line **118**, both of which are in fluid communication with hydrogenotrophic methanogenesis bioreactor **120**.

(24) FIG. 1B provides a more detailed illustration of an embodiment of the electrochemical reactor of the system in FIG. 1A. Reactor **112** comprises anode **130** and cathode **132** with resin wafer **134** therebetween. Cation exchange membrane **136** separates anode **130** from resin wafer **134**. Optional porous gas and liquid distributor **150** is positioned at the end of resin wafer **134** where gas and liquid are introduced into the wafer. Bipolar membrane **138** separates resin wafer **134** from cathode **132**. Reactor **112** includes a gas vent line **114**, a liquid transfer line **116** in fluid communication with hydrogenotrophic methanogenesis bioreactor **120** for transferring aqueous bicarbonate into bioreactor **120** during use. Hydrogen transfer line **118** provides fluid communication between bioreactor **120** and region **133** between cathode **132** and bipolar membrane **138** for transferring hydrogen generated at cathode **132** into bioreactor **120** during use. In some preferred embodiments gas and liquid distributor **150** is a microporous polymeric foam, which distributes the entering gas

and liquid evenly across the resin wafer **134** to improve the uniformity of flow within the wafer.

(25) FIG. **2** is a schematic illustration of an alternative embodiment of an electrochemical reactor, which includes multiple reactor cells connected in series in a stack such that liquid and gas flow sequentially through each reactor cell, and with back to back anodes and cathodes separated by an electrical insulator. Electrochemical reactor **212** comprises anodes **230** and cathodes **232** with resin wafers **234** between the anodes and the cathodes. Cation exchange membranes **236** separates anodes **230** from resin wafers **234**. Bipolar membranes **238** separates resin wafers **234** from cathodes **232**. Each anode **230** is adjacent a cathode **232** of an adjacent reactor cell, with an electrical insulator **240** between the adjacent anodes and cathodes. Reactor **212** includes a gas vent line **214**, a liquid transfer line **216** in fluid communication with hydrogenotrophic methanogenesis bioreactor **220** for transferring aqueous bicarbonate into bioreactor **220** during use. Hydrogen transfer line **218** provides fluid communication between bio reactor **220** and hydrogen collecting regions **233** between cathodes **232** and bipolar membranes **238** for transferring hydrogen generated at cathodes **232** into bioreactor **220** during use. Each reactor cell is in fluid communication with its closest adjacent cell so that the liquid and gas flow serially through each cell with gas and liquid initially entering the first cell in the stack and ultimately existing from the last cell in the stack. In particular the resin wafer **234** of the first cell serially in fluid communication with each subsequent resin wafer **234** by conduits **260**; hydrogen collecting region **233** of the first cell in the stack is serially in fluid communication with each subsequent region **233** by conduits **262**; and hydrogen transfer line **218** is in direct fluid communication with region **233** of the last cell in the stack. Similarly, gas vent **214** and liquid transfer line **216** are in direct fluid communication with the last cell in the stack; while gas transfer lines **210** and **224** are in direct fluid communication with the first cell in the stack. Each resin wafer **234** is in fluid communication with an optional gas and liquid distributor **250** at the end of the resin wafer where liquid and gas enter.

(26) Referring now to FIG. **3**, a flow diagram of a method for generating renewable methane from organic materials is provided. Step **300** involves generating a biogas comprising methane and carbon dioxide by anaerobic degradation of biological material in an anaerobic digester. Step **302** is passing the biogas generated in step **300** through a electrochemical reactor comprising an anode spaced from a cathode by a porous ion exchange resin wafer with a cation exchange membrane between the anode and the resin wafer and a bipolar ion exchange membrane between the cathode and the resin wafer, while applying a voltage across the anode and the cathode to convert carbon dioxide in the biogas to bicarbonate and to generate hydrogen gas by electrochemical water splitting at the cathode. Step **304** is venting and collecting the methane from the biogas that passes through the electrochemical reactor. Step **306** is passing the hydrogen and bicarbonate formed in step **302** into a hydrogenotrophic methanogenesis bioreactor. Step **308** is generating methane from the hydrogen and bicarbonate in the hydrogenotrophic methanogenesis bioreactor. And step **310** is venting and collecting the methane that forms in the hydrogenotrophic methanogenesis bioreactor.

Example 1. Evaluation of Electrochemical Removal of Carbon Dioxide from a Gas Stream in an Electrochemical Reactor

(27) An ERCHD comprising a single electrochemical reactor cell was assembled to demonstrate the concept of simultaneous CO₂ capture and hydrogen production via electrochemical reaction as shown in FIG. **4**. The apparatus in FIG. **4** comprises an anode **430** and a cathode **432**, with a resin wafer **434** therebetween. A cation exchange membrane **436** separates anode **432** from resin wafer **434** and a bipolar membrane **438** separates cathode **432** from resin wafer **434**. Carbon dioxide and aqueous sodium hydroxide are fed into resin wafer **434** through feed line **410** while a voltage is applied across the anode and the cathode. Hydrogen generated in the region between bipolar membrane **438** and cathode **432** is vented and collected at hydrogen transfer line **418**, and aqueous bicarbonate and remaining gaseous carbon dioxide exit resin wafer **434** through outlet **416**.

(28) There are three flow compartments, from right to left in the apparatus off FIG. **4**: (1) An anodic compartment **270** between anode **430** and cation exchange membrane **436**, filled with an

anolyte solution of 10 mM NaOH (2) a CO.sub.2 capture compartment comprising the resin wafer **434** and bipolar membrane **438** filled with 1 mM NaOH and CO.sub.2 gas; and (3) a cathodic compartment **433** adjacent cathode **432** filled with a catholyte of 2.5 wt % Na.sub.2SO.sub.4. (29) During the experiments, the CO.sub.2 content in the gas phase and liquid phase were measured and analyzed in the inlet and outlet of the CO.sub.2 capture compartment. Hydrogen gas produced in the cathodic compartment was also measured. Samples were taken with and without the applied electric field. The results are shown in tabular format in Tables 1, 2, and 3 for three different runs, FIG. 5, and FIG. 6. CO.sub.2 contents in the outlet gas phase (measured as percentage in the air) and in the outlet liquid phase (presented in bicarbonate form and measured as Total Inorganic Carbon (TIC)) were recorded in Tables 1-3.

(30) TABLE-US-00001 TABLE 1 % TIC TIC Sample CO.sub.2* (mg/L) (mM) Samples from CO.sub.2 tank and feed tank — 19.0% 46.15 3.85 Samples no gas or liquid flow — 0.2% — — from flow, no voltage 1 13.1% 58.89 4.91 gas/liquid gas liquid flow, 1.2 V 2 14.1% 63.19 5.27 separator 3 15.4% 63.54 5.30 4 16.2% 69.15 5.76 *excluding broad peaks at 0.7 and 6.7 min

(31) TABLE-US-00002 TABLE 2 % TIC TIC Sample CO.sub.2 (mg/L) (mM) Samples from CO.sub.2 tank and feed tank — 19.0% 38.63 3.22 Samples no gas or liquid flow — 0.4% — — from flow, no voltage 1 10.7% 45.67 3.81 gas/liquid gas liquid flow, 10 V 2 12.3% 47.23 3.94 separator 3 12.6% 50.16 4.18 4 13.0% 50.26 4.19 5 13.0% 51.42 4.29 6 12.8% 54.12 4.51 7 12.9% 57.62 4.80

(32) TABLE-US-00003 TABLE 3 Time TIC TIC Sample (min) % CO.sub.2 (mg/L) (mM) Samples from CO.sub.2 tank & feed — — 20.6% 3.15 0.26 tank Samples no gas or liquid flow 0 0 1.1% — — from flow, no voltage 1 15 12.1% 24.15 2.01 gas/liquid gas liquid flow, 2 30 14.0% 31.07 2.59 separator 1.2 V 3 45 11.2% 29.98 2.50 4 75 10.3% 31.65 2.64 5 105 10.4% 27.90 2.33 6 135 10.9% 27.69 2.31

(33) Low electric field (Table 1, 1.2 V) and high applied field (Table 2 and 3, 10 V) have an impact on the CO.sub.2 capture rate. FIG. 5 shows the CO.sub.2 capture percentage and hydrogen content in the air during the operation. FIG. 6 shows the mass balance of CO.sub.2 capture and hydrogen production during the operation, almost equal molar amounts of CO.sub.2 were captured when the hydrogen also was produced (i.e., when a voltage was applied). These results show that an ERCHD system with simultaneous CO.sub.2 capture and hydrogen production in is more efficient at CO.sub.2 capture than just CO.sub.2 capture without hydrogen production.

(34) All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

(35) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention.

(36) Preferred embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Variations of those preferred embodiments may

become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein. Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

Claims

1. A system for biologically generating methane and removing carbon dioxide therefrom, the system comprising: (a) a primary anaerobic digester adapted and arranged to generate a gaseous mixture comprising methane and carbon dioxide from organic materials; (b) an electrochemical reactor; and (c) a hydrogenotrophic methanogenesis bioreactor adapted and arranged to convert the bicarbonate and hydrogen from the electrochemical reactor to methane; wherein: the electrochemical reactor comprises a plurality reactor cells in a stack with each reactor cell including: an anode; a cathode spaced from the anode; a single porous ion exchange resin wafer between the anode and the cathode, wherein the ion exchange resin wafer consists of a basic ion exchange resin wafer; a cation exchange membrane adjacent to the anode between the anode and the resin wafer, wherein the cation exchange membrane is adjacent to both the anode and the resin wafer; and a bipolar ion exchange membrane between the cathode and the resin wafer, wherein the bipolar exchange membrane is adjacent to both the cathode and the resin wafer; wherein: the cathode of each adjacent reactor cell is separated from the anode of the nearest adjacent cell by an electrical insulator; the reactor cells are in fluid communication with each other in series so that liquid and gas flow through each resin wafer sequentially; and the electrochemical reactor includes a hydrogen transfer line to direct hydrogen gas from the cathodes out of the electrochemical reactor; the electrochemical reactor is adapted and arranged to capture gaseous carbon dioxide within the resin wafer as aqueous bicarbonate, and to electrochemically generate hydrogen gas at the cathodes when an aqueous liquid and a carbon dioxide-containing gas are flowing through the reactor cells, and a voltage is applied across the anode and the cathode of each reactor cell; the primary anaerobic digester includes a gas transfer line in fluid communication with an inlet end of the electrochemical reactor adapted and arranged to receive gas and liquid into pores of the resin wafer; the electrochemical reactor further includes a first gas vent line; an outlet operably connected to a liquid transfer line in fluid communication with the hydrogenotrophic methanogenesis bioreactor; and a hydrogen transfer line providing fluid communication between the hydrogenotrophic methanogenesis bioreactor and a region adjacent the cathode where hydrogen is generated during use; and the hydrogenotrophic methanogenesis bioreactor comprises: a liquid recycle line in fluid communication with the inlet end of the electrochemical reactor adapted and arranged to direct fluid flow through the electrochemical reactor; and a second gas vent line; wherein in use, the electrochemical reactor is filled with an aqueous fluid; the gaseous mixture containing methane and carbon dioxide is formed from the organic materials in the primary anaerobic digester; and the gaseous mixture is transferred from the primary anaerobic digester via the gas transfer line into the inlet end of the electrochemical reactor to flow through the resin wafer while a voltage is applied across the cathode and the anode; the carbon dioxide from the gaseous mixture is electrochemically converted to bicarbonate in the aqueous fluid within electrochemical reactor; hydrogen is electrochemically generated at the cathode; the methane from the gaseous mixture and the aqueous fluid both flow through the resin wafer, and carbon dioxide-depleted methane is vented through the first gas vent line and stored for later use; the aqueous fluid containing the bicarbonate flows through the liquid transfer line into the hydrogenotrophic methanogenesis bioreactor, and the hydrogen generated at the cathode flows through the hydrogen

transfer line into the hydrogenotrophic methanogenesis bioreactor, where the bicarbonate and hydrogen are biologically converted to methane; the methane generated in the hydrogenotrophic methanogenesis bioreactor is vented through the second gas vent line and collected for later use; and aqueous fluid is recycled back into the inlet end of the electrochemical reactor from the hydrogenotrophic methanogenesis bioreactor through the liquid recycle line.

2. The system of claim 1, wherein the primary anaerobic digester comprises: (i) an acidogenesis reactor adapted and arranged to biologically convert organic waste and wastewater to soluble volatile fatty acids (VFA), methane, carbon dioxide, and hydrogen; and (ii) an acetoclastic methanogenesis reactor adapted and arranged to biologically convert the VFA to carbon dioxide and methane; wherein the gas transfer line is in fluid communication with both the acidogenesis reactor and the acetoclastic methanogenesis reactor; and the acidogenesis reactor is in fluid communication with the acetoclastic methanogenesis reactor via a liquid feed line; and wherein in use, VFA, carbon dioxide, hydrogen and methane are generated in the acidogenesis reactor; VFA-containing liquid from the acidogenesis reactor is fed into the acetoclastic methanogenesis reactor via the liquid feed line; methane and carbon dioxide are generated in the acetoclastic methanogenesis reactor, and carbon dioxide, methane, and hydrogen are transferred from the acidogenesis reactor and the acetoclastic methanogenesis reactor into the inlet end of electrochemical reactor through the gas transfer line.

3. The system of claim 1, wherein each of the reactor cells comprises at least one porous gas and liquid flow distributor in contact with the resin wafer thereof; the flow distributor being adapted and arranged to laterally distribute gas and liquid within pores thereof prior to entering the resin wafer of the electrochemical reactor.

4. The system of claim 3, wherein the at least one porous gas and liquid flow distributor includes interconnected pores having an average pore size in the range of about 100 to about 600 micrometers, and is adapted and arranged to laterally distribute the liquid and gas bubbles having an average diameter in the range of about 100 to about 600 micrometers throughout the flow distributor and into the resin wafer of the electrochemical reactor.

5. The system of claim 1, wherein the electrochemical reactor comprises at least one porous gas and liquid flow distributor in contact with at least one of the resin wafers in the stack; and the at least one porous gas and liquid flow distributor is adapted and arranged to distribute gas within the liquid flowing through the resin wafers of the electrochemical reactor.

6. The system of claim 5, wherein the at least one porous gas and liquid flow distributor includes interconnected pores having an average pore size in the range of about 100 to about 600 micrometers, and is adapted and arranged to laterally distribute the liquid and gas bubbles having an average diameter in the range of about 100 to about 600 micrometers throughout the flow distributor and into the resin wafers of the electrochemical reactor.

7. The system of claim 1, wherein each wafer has a thickness in the range of about 1 to about 20 mm.

8. The system of claim 1, wherein the electrochemical reactor comprises 2 to about 50 reactor cells.

9. An electrochemical reactor for capturing carbon dioxide as bicarbonate and generating hydrogen; the reactor comprising a plurality reactor cells in a stack with each reactor cell including: (a) an anode; (b) a cathode spaced from the anode; (c) a single porous ion exchange resin wafer between the anode and the cathode, wherein the ion exchange resin wafer consists of a basic ion exchange resin wafer; (d) a cation exchange membrane between the anode and the resin wafer, wherein the cation exchange membrane is adjacent to both the anode and the resin wafer; and (e) a bipolar ion exchange membrane between the cathode and the resin wafer, wherein the bipolar exchange membrane is adjacent to both the cathode and the resin wafer; wherein: (A) the cathode of each adjacent reactor cell is separated from the anode of the nearest adjacent cell by an electrical insulator; (B) the reactor cells are in fluid communication with each other in series so that liquid and gas flow through each resin wafer sequentially; and (C) the electrochemical reactor includes a

hydrogen transfer line to direct hydrogen gas from the cathodes out of the electrochemical reactor; and the electrochemical reactor is adapted and arranged to capture gaseous carbon dioxide within the resin wafer as aqueous bicarbonate, and to electrochemically generate hydrogen gas at the cathodes when an aqueous liquid and a carbon dioxide-containing gas are flowing through the reactor cells, and a voltage is applied across the anode and the cathode of each reactor cell.

10. The electrochemical reactor of claim 9, further comprising a gas vent line for venting a carbon dioxide depleted gas from the resin wafers of the electrochemical reactor.

11. The electrochemical reactor of claim 9, wherein the electrochemical reactor comprises at least one porous gas and liquid flow distributor in contact with at least one of the resin wafers in the stack; and the at least one porous gas and liquid flow distributor is adapted and arranged to distribute gas within the liquid flowing through the resin wafers of the electrochemical reactor.

12. The electrochemical reactor of claim 11, wherein the at least one porous gas and liquid flow distributor includes interconnected pores having an average pore size in the range of about 100 to about 600 micrometers, and is adapted and arranged to laterally distribute the liquid and gas bubbles having an average diameter in the range of about 100 to about 600 micrometers throughout the distributor and into the resin wafers of the electrochemical reactor.

13. The electrochemical reactor of claim 9, wherein each wafer has a thickness in the range of about 1 to about 20 mm.

14. The electrochemical reactor of claim 9, wherein the electrochemical reactor comprises 2 to about 50 reactor cells.

15. A method for biologically generating carbon dioxide-depleted biogas comprising the steps of: (a) generating a biogas comprising methane and carbon dioxide by anaerobic degradation of biological material in an anaerobic digester; (b) passing the biogas generated in step (a) through the electrochemical reactor of claim 10, while applying a voltage across the anode and the cathode to convert carbon dioxide in the biogas to bicarbonate and to generate hydrogen gas by electrochemical water splitting at the cathode; (c) venting and collecting the methane from the biogas that passes through the electrochemical reactor; (d) passing the hydrogen and bicarbonate formed in step (b) into a hydrogenotrophic methanogenesis bioreactor; (e) generating methane from the hydrogen and bicarbonate in the hydrogenotrophic methanogenesis bioreactor; and (f) venting and collecting the methane that forms in the hydrogenotrophic methanogenesis bioreactor.

16. The method of claim 15, wherein the anaerobic digester comprises: (i) an acidogenesis reactor adapted and arranged to biologically convert organic waste and wastewater to soluble volatile fatty acids (VFA), methane, carbon dioxide, and hydrogen; and (ii) an acetoclastic methanogenesis reactor adapted and arranged to biologically convert the VFA to carbon dioxide and methane; wherein in use, VFA, carbon dioxide, hydrogen and methane are generated in the acidogenesis reactor; VFA-containing liquid from the acidogenesis reactor is fed into the acetoclastic methanogenesis reactor; methane and carbon dioxide are generated in the acetoclastic methanogenesis reactor, and carbon dioxide, methane, and hydrogen are transferred from the acidogenesis reactor and the acetoclastic methanogenesis reactor into electrochemical reactor.

17. The method of claim 15, wherein the electrochemical reactor comprises at least one porous gas and liquid flow distributor in contact with at least one of the resin wafers in the stack; and the at least one gas distributor is adapted and arranged to distribute gas within the liquid flowing through the resin wafers of the electrochemical reactor.

18. The method of claim 17, wherein the at least one porous gas and liquid flow distributor includes interconnected pores having an average pore size in the range of about 100 to about 600 micrometers, and is adapted and arranged to laterally distribute the liquid and gas bubbles having an average diameter in the range of about 100 to about 600 micrometers throughout the flow distributor and into the resin wafers of the electrochemical reactor.

19. The apparatus of claim 9 wherein each electrical insulator comprises a non-conductive polymer.
